Structured Synthetic Supervision and Metric Pitfalls in Ultra-Low-Resource Tangut Short-Text Translation

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Abstract

Tangut translation is an extreme low-resource problem: in the current repository snapshot, only 491 real Tangut-Chinese pairs are available, with 400/41/50 train-dev-test examples. The task is also structurally unusual because many targets are short, title-like Chinese strings rather than ordinary modern sentences. We evaluate three families of methods in this setting: inference-only prompting, synthetic-data supervised fine-tuning (SFT), and direct preference optimization (DPO). The strongest current result comes from a structured multitask SFT design that exposes intermediate alignment structure during training; it achieves the best non-reference chrF++ score (25.99), the highest exact-match count (5/50), and the best overall score (2.06/5) under a new reference-aware judge. We further show that several apparently reasonable reference-free metrics are misaligned with the task: the gold human reference itself receives only moderate dictionary coverage and extremely poor perplexity under a modern-Chinese language model, despite achieving perfect chrF++ and perfect reference-aware judge scores. Finally, we report a negative result for reward-based alignment: DPO trained with a lexical-coverage-plus-perplexity reward degrades reference-aligned performance and introduces output contamination. These findings suggest that, for ultra-low-resource Tangut short-text translation, explicit structural supervision is currently more reliable than preference optimization, and evaluation must be anchored to reference-aware metrics rather than generic fluency proxies.

1 Introduction

Machine translation for historical and ultra-low-resource languages is difficult for two separate reasons. First, parallel data is often scarce or nearly absent. Second, the outputs that matter to scholars are not always long, natural modern sentences: they may be titles, glosses, liturgical fragments, or compact bibliographic strings. Recent work on ancient-language NLP and historical-text translation has made this setting more visible, but it has also shown that data scarcity and evaluation instability are especially severe outside modern benchmark conditions [1, 2, 3]. Tangut is a canonical example. The script is historically important, but usable parallel Tangut-Chinese resources remain tiny. In the current repository snapshot, we only have 491 real Tangut-Chinese pairs.

This small-data regime creates a practical question: when full supervised translation is impossible, what kind of signal is most helpful? A natural first option is dictionary-augmented prompting, in the spirit of lexicon-guided translation. A second option is to synthesize pseudo-parallel training data from dictionary knowledge and monolingual Chinese. A third option is to add preference optimization on top of an SFT base [5, 4, 7]. The repository already contains experiments spanning all three options, which makes it possible to write a focused empirical paper from the existing evidence.

Our main finding is that structured synthetic supervision currently works best. Among all non-reference systems, the strongest model is a multitask SFT variant that asks the model to expose intermediate alignment structure during training. This system is not best because it maximizes dictionary coverage or because it looks best under a generic Chinese language model. It is best because it most closely matches the reference outputs in shape and content: it has the highest current chrF++, the highest exact-match rate, and the best title-suffix preservation among learned systems.

The repository also reveals two important failure modes. First, reference-free metrics can be badly misleading. The human reference itself is penalized by lexical coverage and perplexity, which means those metrics cannot serve as primary model-selection criteria for this task. Second, preference optimization is unstable under the current reward design. The DPO model underperforms its SFT base and frequently produces contaminated outputs, suggesting a mismatch between the reward and the actual task objective.

The current draft therefore makes three contributions:

1. an empirical comparison of prompting, synthetic SFT, and DPO for Tangut short-text translation;

2. a positive method result showing that structured synthetic supervision is the strongest current approach;

3. a reference-aware evaluation and alignment analysis showing why reference-free metrics and the current DPO reward can fail.

The rest of the paper situates the task against related work, describes the experimental setting and the three method families, and then separates the positive result from the metric and alignment failure analysis that makes the result interpretable.

2 Related Work

Ancient and historical language processing. Recent surveys argue that ancient-language NLP differs from mainstream NLP not just because data are scarce, but because scripts, editorial conventions, and annotation practices are unstable across corpora [1]. Shared tasks on ancient-language translation have likewise emphasized that even when translation is the end goal, the available supervision is often tiny and domain-specific [2]. Work on historical document modernization faces related sparsity and spelling-shift issues, but often targets orthographic or lexical modernization rather than cross-script title reconstruction [3]. Our Tangut setting is even narrower: only 491 real pairs are available, and the evaluation set is dominated by compact title-like strings.

Synthetic supervision and lexical grounding. Low-resource machine translation often relies on synthetic parallel data built from monolingual corpora, with back-translation as the canonical example [4]. A separate line of work injects lexical knowledge more directly, either by folding bilingual lexicons into the model or by constraining decoding to respect required terminology [5, 6]. Our prompting and SFT baselines sit at the intersection of these ideas: they exploit dictionary knowledge and synthetic supervision, but the central question is whether stronger gains come from input purity, approximate semantic preservation, or explicit alignment structure.

Preference optimization. Direct Preference Optimization (DPO) has become a standard offline objective for aligning generation models without a separate reward-model-plus-RL loop [7]. In many open-ended generation settings, DPO is attractive because it can transfer weak preference signals into better outputs with a relatively simple objective. Our results show that this promise does not transfer automatically to ultra-low-resource translation when the chosen/rejected signal is built from partially misaligned automatic metrics.

Evaluation for non-standard translation outputs. chrF++ is widely used in machine translation because character-level overlap is often more stable than token-level overlap under morphological or orthographic variation [8]. Learned evaluation and quality-estimation frameworks such as COMET and OpenKiwi push beyond surface overlap by predicting translation quality from richer signals [9, 10]. Our findings add a complementary caution: in short historical title translation, apparently reasonable reference-free metrics can still mis-rank the gold reference itself. This is why the current paper treats reference-aware evidence as primary and reference-free metrics as diagnostic.

3 Task Setting

The real Tangut data in this repository contains 491 pairs. The current split is 400 training examples, 41 development examples, and 50 test examples. Unlike ordinary machine-translation benchmarks, the targets are short and often title-like. Many examples resemble book titles, sutra titles, or compact scholastic expressions. This matters because title reconstruction rewards precise lexical choices and compact formatting, while a modern Chinese fluency model may incorrectly treat those outputs as unnatural.

Because the real dataset is tiny, the system uses synthetic data to train translation models. For each SFT branch, 50,000 synthetic Tangut-Chinese pairs are produced from monolingual classical Chinese and dictionary knowledge, then combined with the 400 real training examples to form a 54,000-example training set. All learned models use the same instruction-tuned 7B backbone.

We evaluate with two classes of metrics. The repository's existing metrics are lexical coverage, perplexity, chrF++, and a dictionary-conditioned LLM judge. chrF++ is a reasonable overlap anchor for this setting because character-based scoring is relatively stable under orthographic variation [8]. Learned MT evaluation and quality-estimation systems can capture richer signals than surface overlap [9, 10], but in this repository the available reference-free signals are only partially aligned with the task. For that reason, this draft does not treat all metrics equally: chrF++, exact match, contamination, suffix preservation, and the new reference-aware judge are primary evidence, while lexical coverage, perplexity, and the legacy reference-free judge are reported as diagnostics.

4 Methods

4.1 Inference-only baselines

The first family contains no weight updates. `baseline1` directly prompts the base model with the Tangut string. `baseline2` augments the prompt with dictionary glosses. `baseline2_1_cot` further restructures the prompt into a three-step process intended to encourage alignment, literal drafting, and rewriting. This family is closest in spirit to lexicon-augmented or terminology-aware MT, except that the lexicon is injected through prompting rather than through decoding constraints or model-side features [5, 6].

These methods are important because they show what can be achieved from symbol injection alone. In the current results, dictionary prompting is substantially better than zero-shot generation, but it often expands compact titles into explanatory paraphrases rather than reconstructing the reference form.

4.2 Synthetic SFT baselines

The second family uses synthetic training data.

- `baseline3` uses mixed Tangut-Chinese inputs.
- `baseline3_1_unk` enforces pure Tangut inputs by replacing dictionary misses with `UNK`.
- `baseline3_3_semantic` also enforces pure inputs, but replaces misses through vector-space semantic projection instead of `UNK`.
- `baseline3_2_multitask` introduces a structured target containing explicit alignment scaffolding.

The central design question is whether the model benefits more from input purity, approximate semantic preservation, or explicit alignment structure. At a high level, this family plays the role that synthetic parallel data often plays in low-resource MT: it converts monolingual or weakly aligned resources into supervised signals that a translation model can absorb [4]. The current evidence favors explicit alignment structure.

4.3 Preference optimization

The final family applies DPO to an SFT base using reward-defined chosen/rejected pairs [7]. In the current repository, the reward combines lexical coverage with a penalty on log perplexity. The pipeline first selects the best SFT base among the mixed, `UNK`, and semantic variants using chrF++, then builds preference pairs from that base. In the current snapshot, the selected base is the `UNK` model.

This design is informative even though it does not currently succeed. Because the reward is partly built from misaligned signals, DPO becomes a useful negative-result case study.

5 Main Results

Table 1 summarizes the repository metrics. The central pattern is straightforward. Zero-shot generation fails. Dictionary prompting is much stronger, especially on lexical coverage. Synthetic SFT improves reference alignment substantially. Among the SFT family, the multitask structured variant is best. DPO does not improve on its SFT base.

The most important comparison is between `baseline3_1_unk`, `baseline3_2_multitask`, and `final_v2`. The `UNK` model already demonstrates that pure-input training is helpful. The multitask model improves further, reaching chrF++ 25.99 and 5 exact matches out of 50 test items. By contrast, `final_v2` falls back to chrF++ 19.39 and only 2 exact matches, while also showing visible contamination in some outputs.

This result suggests that, in the current Tangut setting, preference optimization is not the easiest way to exploit scarce supervision. A better strategy is to inject stronger structure during supervised training.

6 Analysis

6.1 Reference-aligned diagnostics

Repository-level metrics alone do not fully explain why `baseline3_2_multitask` is the best current model. Additional diagnostics from stored predictions make the result clearer. The multitask

Method	Lex ↑	PPL ↓	chrF++ ↑	Sem ↑	Flu ↑
baseline1	0.1165	20.21	0.19	1.00	1.56
baseline2	0.8187	4474.44	6.13	2.30	2.40
baseline2_1_cot	0.8723	18119.16	7.59	2.30	1.88
baseline3	0.3192	298684.91	17.85	1.42	1.88
baseline3_1_unk	0.3514	11031.18	22.11	1.66	1.96
baseline3_2_multitask	0.4839	3322.93	**25.99**	1.70	2.26
baseline3_3_semantic	0.3723	17064.38	17.03	1.46	1.84
final_v2	0.4224	314577.16	19.39	1.52	1.76
human_reference	0.5733	1646139.34	100.00	2.24	2.34

Table 1: Repository-level metrics on the 50-example Tangut test set. The best non-reference result is `baseline3_2_multitask`.

Method	Exact Match	Contam. Rate	Len. Ratio	Suffix Preserve
baseline2	0/50	0.00	1.83	3/27
baseline3_1_unk	3/50	0.06	1.05	17/27
baseline3_2_multitask	**5/50**	**0.02**	1.10	**19/27**
final_v2	2/50	0.18	1.88	18/27
human_reference	50/50	0.00	1.00	27/27

Table 2: Reference-aligned diagnostics derived from stored predictions. "Contam." denotes mixed-script or artifact contamination.

model produces the best exact-match count (5/50), the best title-suffix preservation among learned systems (19/27), and a near-reference output length ratio (1.10). These properties matter for a title-like task.

By contrast, the dictionary-prompting baseline often produces outputs that are too long and too explanatory. Its average output is about 1.83 times the reference length. This helps lexical coverage but hurts exact title reconstruction.

6.2 Reference-aware judging confirms the main ranking

The legacy LLM judge in the repository is reference-free, so we reran evaluation with a stricter judge that sees the Tangut input, the gold reference, and the candidate output simultaneously. This rerun serves two purposes. First, it checks whether the human reference is correctly ranked when the judge has the information it should need. Second, it tests whether the main system ranking survives once the judge can penalize explanatory overgeneration directly.

Table 3 confirms the current paper narrative. The human reference receives perfect 5.0 scores on all dimensions. Among non-reference systems, `baseline3_2_multitask` is best on reference agreement (1.92), source faithfulness (2.08), and overall quality (2.06). The `UNK` model is slightly higher on title-style fitness (4.50 vs. 4.24), which suggests that it is somewhat more conservative about staying short, but that conservatism does not translate into better content recovery.

The subset breakdown makes the task boundary sharper. On the 27 reference titles containing common title suffixes, `baseline3_2_multitask` raises exact match from 7.4% to 14.8% relative to `baseline3_1_unk`, increases mean overall judge score from 1.93 to 2.30, and reduces contamination from 11.1% to 0.0. On the 23 non-suffix items, the advantage nearly disappears under the judge, which is consistent with the broader claim that the current repository most strongly supports title-like Tangut translation rather than general sentence translation. At the example level,

Method	Ref. Agr.	Src. Faith.	Title Style	Overall
baseline2	1.50	1.52	2.10	1.50
baseline3_1_unk	1.70	1.80	**4.50**	1.76
baseline3_2_multitask	**1.92**	**2.08**	4.24	**2.06**
final_v2	1.64	1.78	3.72	1.72
human_reference	5.00	5.00	5.00	5.00

Table 3: Reference-aware judge scores on a 1–5 scale. The judge sees the Tangut input, the gold reference, and the candidate output. `baseline3_2_multitask` is the strongest non-reference system on agreement, faithfulness, and overall quality.

`baseline3_2_multitask` beats `baseline3_1_unk` on the judge's overall score for 16 test items, loses on 7, and ties on 27; against `final_v2`, it leads 21–8 with 21 ties.

6.3 Human-reference anchor

One of the strongest pieces of evidence in the repository is the `human_reference` experiment. The gold reference receives perfect chrF++ and perfect reference-aware judge scores, as expected, but only moderate lexical coverage and catastrophic perplexity. This reveals a core evaluation issue: the current lexical coverage metric measures overlap with dictionary-derived semantics rather than agreement with the gold title, while the perplexity model penalizes compact title style and historical transliterations.

The implication is methodological, not cosmetic. If we were to select models purely by lexical coverage or perplexity, we would risk preferring systems that paraphrase or overgenerate instead of systems that best reconstruct the true title. Therefore, in the paper, lexical coverage and perplexity should be reported as supporting diagnostics, not as the primary decision criteria.

6.4 Why the current DPO fails

The DPO result is not a random bad run. The preference data itself is imperfect. The repository contains 3,867 preference pairs, including a small number of duplicate or invalid entries. More importantly, a quick audit against the original synthetic targets shows that the reward-chosen candidate is closer to gold only about 71% of the time under a simple similarity proxy, and the low-gap pairs are much noisier. This means that the reward is informative but far from clean.

The current selector also excludes the strongest multitask model from the DPO-base search. As a result, the paper should avoid claiming that DPO does not work for Tangut in general. A narrower and defensible statement is: under the current reward design and current base-selection policy, DPO is unstable and underperforms structured SFT.

7 Discussion and Limitations

The paper should be explicit about scope. The current dataset supports Tangut short-text translation, especially title-like strings, not general open-domain Tangut sentence translation. The new subset analysis strengthens that boundary: the main gains from `baseline3_2_multitask` are concentrated on reference titles with canonical suffix patterns, while the judge advantage is much smaller on the non-suffix subset. That scope is still meaningful: many historically important Tangut materials appear precisely in compact catalog, title, or scholastic forms, and low-resource methods for such data are valuable.

Audit item	Value
Total preference pairs	3867
Duplicate chosen/rejected pairs	3
Non-finite reward entries	3
Mean reward gap	0.2396
Chosen closer to gold proxy	71.3%
Chosen closer on low-gap pairs (≤ 0.10)	62.3%
Pairs kept with gap ≥ 0.20	1996
Pairs kept with gap ≥ 0.40	498

Table 4: Lightweight audit of the current DPO pair construction pipeline. Lower-gap pairs are noticeably noisier.

The paper should also be honest about absolute difficulty. Even the strongest learned system reaches only 2.06/5 under the reference-aware judge, so the task is far from solved. The result is therefore not that Tangut title translation is easy, but that structured supervision is currently the least fragile option among the methods already implemented in the repository.

Two limitations matter most for future work. First, the real dataset is extremely small, so the paper cannot claim stable behavior beyond the current narrow domain. Second, the DPO conclusion is intentionally narrow: the current pipeline uses a noisy reward, low-gap preference pairs, and a base-selector that excludes the strongest multitask model. The most useful follow-up experiment is therefore not "more DPO" in the abstract, but a multitask-base DPO control or a stricter gap-filtered DPO rerun.

8 Conclusion

The current repository already supports a coherent paper. Structured multitask synthetic supervision is the strongest method in the snapshot. Pure-input training is helpful, but noisy semantic projection does not yet outperform the simpler `UNK` strategy. A new reference-aware judge confirms that the multitask model is the best non-reference system on agreement, faithfulness, and overall quality, while the human reference is correctly scored as perfect. Several reference-free metrics remain misaligned enough to mis-rank the human reference, and DPO, under the current reward and base-selection design, is unstable and best treated as a negative result. Together, these findings argue for a simple practical lesson: in ultra-low-resource Tangut short-text translation, explicit structural supervision is currently more reliable than reward-based alignment, and evaluation must stay anchored to reference-aware evidence.

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